open file list → Which files are open

device list → What devices are open.

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Process State

Process state is what the process is currently doing, what condition is it present now.

There are different Process state:

(i) New state: (creation)

It is the creation of a process, when a program is taken & loaded into Memory, it is New state.

→ Create process, Assign PID
→ Creates & init PCB
→ Prepare process to push in Ready Queue

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(ii) Ready state (Ready queue)

The process is residing in the Ready queue, Ready to be executed when it time comes, ie, when CPU is available.

(iii) Running state

Process enters Running state, ie, executed when CPU executes its instruction.
Process do not stay in the Running state till end, in b/w, it will go out for different reasons.

(ex) I/O request → Ex) If the process needs ready file, need to wait.

Time quanta expire → OS scheduler takes CPU away from it

(iv) Waiting state

When the process leaves the Running

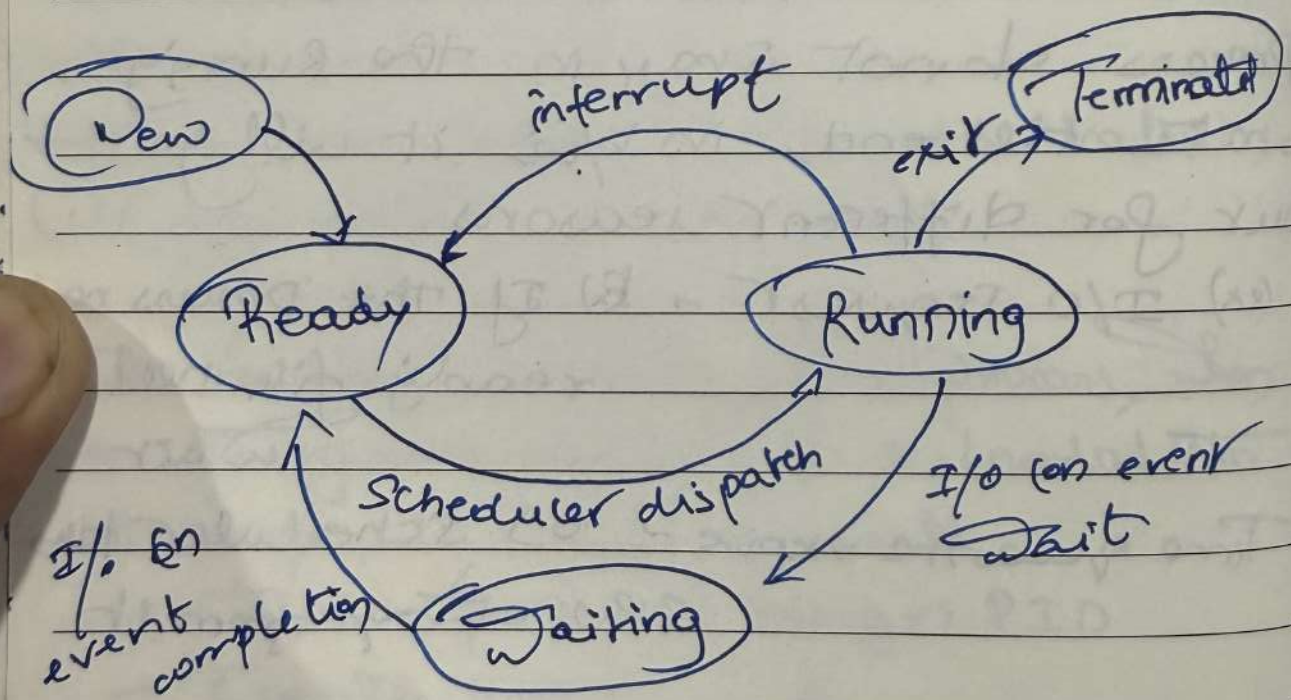
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state, it goes into the waiting state, then eventually enters the Ready queue again.

(v) Terminated:

Execution has been finished & has been stopped by OS



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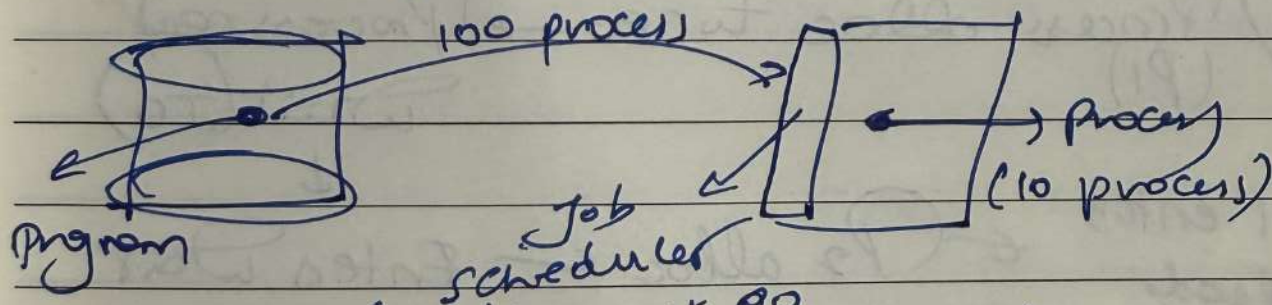
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Types of Schedulers:

(i) Job Scheduler:

→ Also known as Long Term scheduler.

→ It takes care of admitting the program from disk to the Memory by checking whether it's required / not required.



∴ process → Allocating to Ready Queue

(Done by Job scheduler)

(ii) CPU scheduler:

→ Also known as Short Term Scheduler (STS)

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→ Takes care of allocating CPU for executing the process, ie,

Ready state → Running state

It has to be short term, since, it frequency of use will be high.

Ex) Process (P1) alloc to CPU → Process read wait(IO)

P1 enters ready queue again ← P2 alloc ← Enters wait

↓ ... keeps going continuously until all are executed.

Degree of Multiprogramming:

→ Number of processes that could be in Ready queue @ a time.

→ Handled by Job scheduler (LTS).